How WindFlow works

Oct 29, 2026 · @Bijay Karki

WindFlow blows wind through and around your building on the graphics card and reports, room by room, how fresh the air is. It is a design-comparison tool: it ranks options against each other on the same model. It does not certify a building.

From model to report in six steps

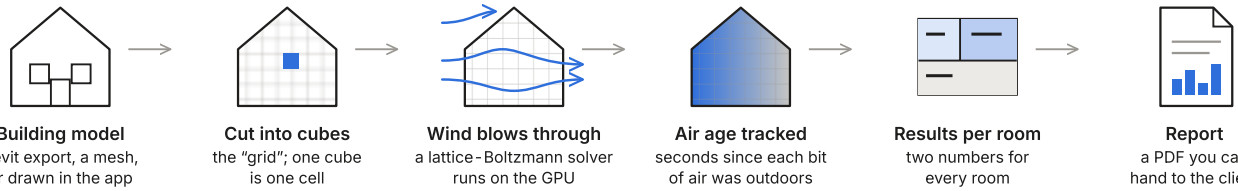


Figure 1 · The pipeline. Blue is fresh outdoor air, grey is air that has been indoors a while; the same colour code is used in every figure below.

Two numbers for every room

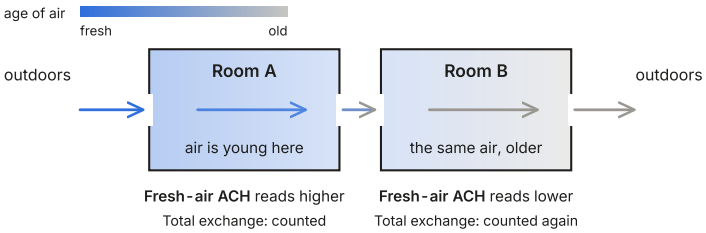


Figure 2 · Two rooms in series. The air that reaches Room B has already aged in Room A, so B's fresh-air number is lower even though the same air passes through both.

Fresh-air ACH

Fresh-air ACH = 3600 ÷ mean age of air (s)

How many times an hour the room's air is replaced by air that is *effectively* fresh. Air arriving through other rooms counts at its real age.

Total exchange

All air entering the room per hour, including air that came from other rooms. It is a measure of how much air moves, **not** a ventilation rate.

Read the two together. A room can have a lot of air moving through it and still little fresh air.

What the numbers are good for

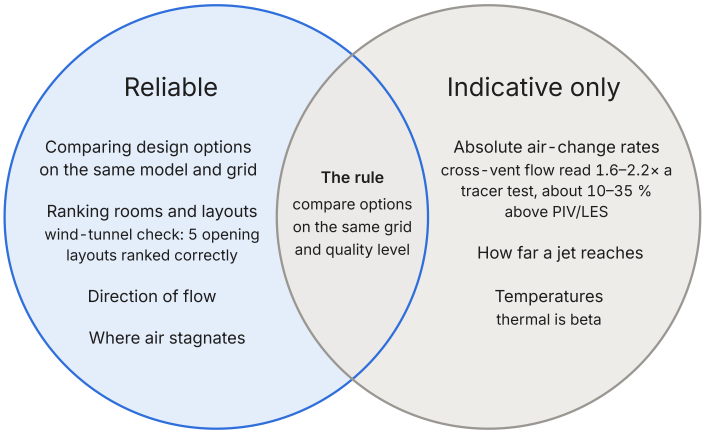


Figure 3 · Trust the ranking, not the absolute number. Changing the cell size alone can move one room by up to about 35 % and the whole building by about 20 %, so never compare results from different grids.

Why a window's size in cubes matters

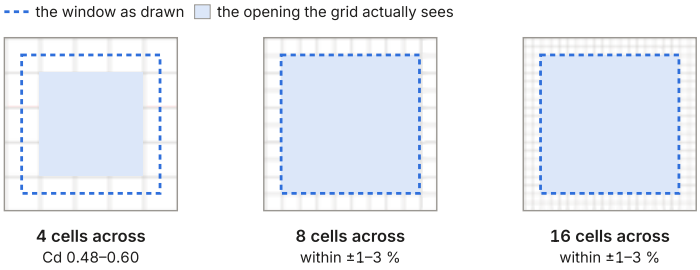


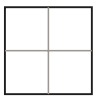
Figure 4 · The same window on three grids. On a coarse grid the window edge lands between cubes, so part of the opening is lost and the result shifts with where the grid happens to fall.

A sharp-edged window lets air through with a discharge coefficient of about **Cd ≈ 0.61**. The grid has to reproduce that.

Cells across	What you get
about 4	Cd 0.48–0.60: flow up to about 20 % low, and the answer depends on exactly where the grid falls
8–16	within about ±1–3 % of the right flow

Finer grids make openings more accurate, but each step finer is much slower. That trade-off is what the quality levels below choose for you.

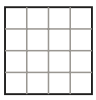
Quality levels pick one and stay on it



Preview

seconds to a minute

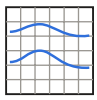
Coarse cubes. Shows the direction of flow only.



Standard

about 10 min

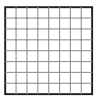
About 170 mm cubes on a typical house. Fine detail settles in 30–50 min.



High

5–30 min

Gusty wind, as on a real site, not a steady breeze.



Ultra

6–24 h

The finest grid your GPU fits, aiming at 16 cubes across the narrowest window. Saves progress and can resume. New; not separately validated.

Figure 5 · Each level cuts the building into smaller cubes than the one before. The numbers change with the cube size, so **never compare numbers across levels**.

What to keep in mind



The ground is smooth
Wind near the ground is too weak: about 54 % under the Eurocode profile at 1 m.



The compact outdoor area crowds the building
Flows read about 11–14 % high. Use **Full** for a final check.



Walls do not conduct heat
No U-values; walls, floors and glazing are adiabatic.



Small heat sources under-stratify
Warm air from a small source does not layer as strongly as it would in reality.



Imported meshes read higher
Total exchange is 37–45 % higher than the same building drawn in the app. Compare imported with imported.



Not yet checked against a measured building
Benchmarks are lab cases and published studies, not a real occupied building.

Full method, equations and every benchmark: WindFlow Technical Reference.

WindFlow is a design-stage tool. Its results are estimates with stated uncertainty, not a substitute for a ventilation engineer or a code-compliance calculation.